System Design Document (SDD)

Smart Plot Investment Platform (SPIP)

1. Project Overview

The **Smart Plot Investment Platform (SPIP)** is a PropTech solution that digitizes the plotted real estate investment process.

The platform enables builders to upload projects, manage plots, and allows investors to browse plots, evaluate legal compliance, and analyze investment opportunities.

Primary Objectives

• Digitize plot buying process
• Provide transparent legal compliance information
• Enable online plot booking
• Provide investment analytics and ROI insights

2. System Actors

Actor	Description
Builder	Creates and manages projects and plots
Investor	Searches plots and invests
Admin	Manages system verification and compliance

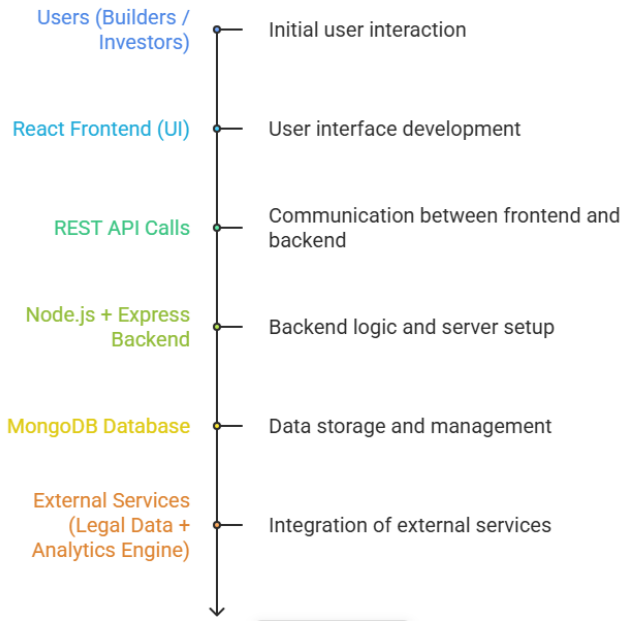
3. Modules Covered

Module	Name	Purpose
Module 2	Builder & Project Management	Manage builder projects and plot inventory
Module 3	Plot Marketplace & Booking	Allow investors to explore and book plots
Module 4	Legal & Compliance Intelligence	Display legal verification data
Module 6	Analytics & Investment Intelligence	Provide investment insights

4. Technology Stack

Layer	Technology
Frontend	React
Backend	Node.js + Express
Database	MongoDB
Authentication	JWT
Testing	Postman
Version Control	GitHub
Deployment	Vercel / Render

5. High Level System Architecture (HLD)



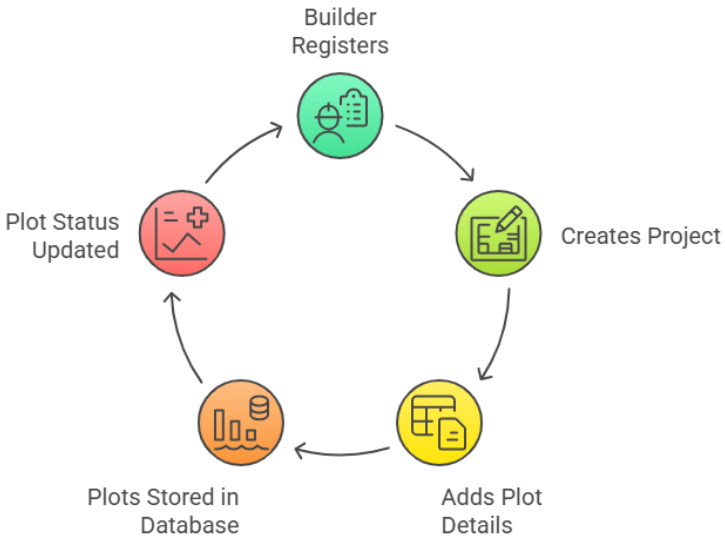
6. Module Design (LLD)

Module 2 — Builder & Project Management

Features

• Builder registration
• Project creation
• Plot inventory management
• Dynamic pricing updates
• Plot status management

Workflow



Main Entities

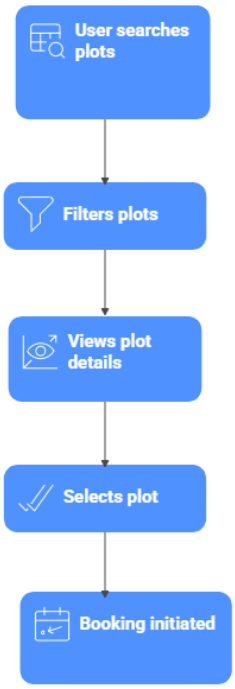
Entity	Description
Builder	Real estate developer
Project	Housing layout
Plot	Individual land unit

Module 3 — Plot Marketplace & Booking

Features

• Interactive plot listing
• Advanced search filters
• Plot comparison
• Real-time availability tracking
• Plot booking functionality

Booking Workflow

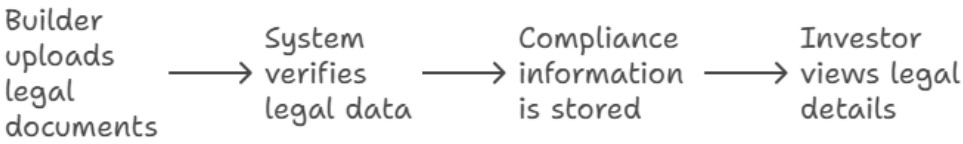


Module 4 — Legal & Compliance Intelligence

Features

• Display Katha classification
• RERA registration tracking
• DC conversion status
• Encumbrance document storage
• Legal risk score generation

Compliance Workflow



Example Legal Data

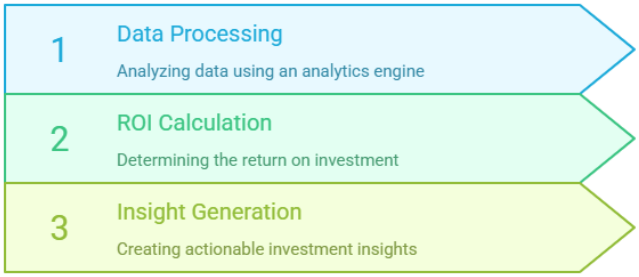
Field	Description
Katha Type	A / B / NA
RERA ID	Registration number
DC Conversion	Land conversion approval
Legal Risk Score	Compliance rating

Module 6 — Analytics & Investment Intelligence

Features

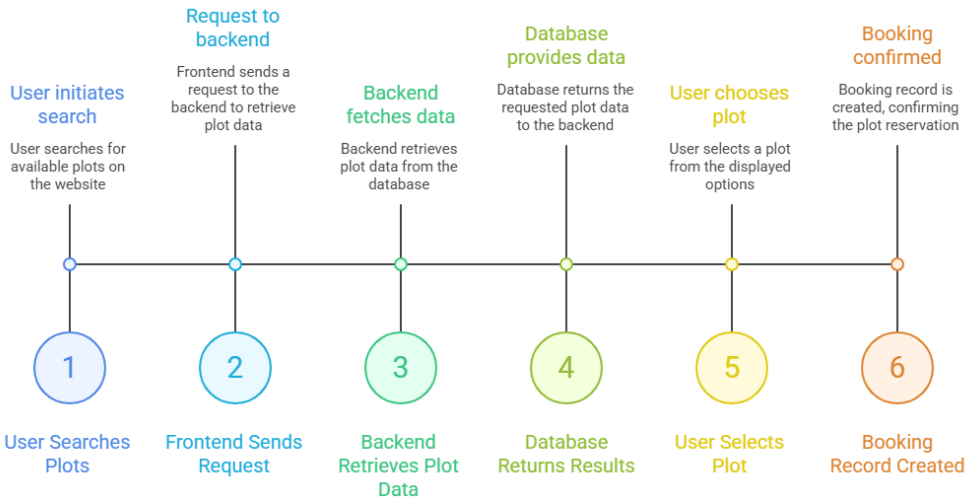
• Price per square foot calculation
• ROI estimation calculator
• Historical price trends
• Area demand heatmap
• Investment recommendation engine

Analytics Flow



7. Data Flow

Plot Investment Data Flow



8. Database Design

Core Collections

Collection	Description
Users	Builder and investor accounts
Projects	Builder projects
Plots	Plot inventory
Legal Documents	Compliance documents
Investments	Plot bookings
Analytics	Investment statistics

ER Diagram

```
Users
  │
  ├──────< Projects
  │            │
  │            └──────< Plots
  │
  ├──────< Investments
  │
  └──────< LegalDocuments
```

9. API Design

Authentication APIs

Method	Endpoint	Purpose
POST	/api/auth/signup	Register user
POST	/api/auth/login	Login user

Builder APIs

Method	Endpoint	Purpose
POST	/api/projects	Create project
GET	/api/projects	Fetch projects

Plot APIs

Method	Endpoint	Purpose
GET	/api/plots	Fetch plot list
GET	/api/plots/:id	Plot details

Legal APIs

Method	Endpoint	Purpose
GET	/api/legal/:plotId	Fetch compliance data

Analytics APIs

Method	Endpoint	Purpose
GET	/api/analytics/roi	ROI calculation
GET	/api/analytics/demand	Area demand

10. Project File Structure

Backend

```
backend
 ├── controllers
 ├── models
 ├── routes
 ├── services
 ├── middleware
 ├── config
 └── utils
```

Frontend

```
src
 ├── components
 ├── pages
 ├── services
 ├── hooks
 └── utils
```

11. Security Design

Security Feature	Implementation
Password encryption	bcrypt
Authentication	JWT
Authorization	Role-based access
API validation	Middleware validation

12. Deployment Architecture

Deployment Platforms

Component	Platform
Frontend	Vercel/Render
Backend	Render/Vercel
Database	MongoDB Atlas

13. Non Functional Requirements

Category	Description
Performance	Support multiple users simultaneously
Scalability	Cloud-ready architecture
Reliability	Secure data storage
Maintainability	Modular service architecture

14. Diagrams

1. Use Case Diagram (Actors vs System Features)

This diagram shows **how different users interact with the system**.

```
                +-----------------------------------+
                |   Smart Plot Investment Platform |
                +-----------------------------------+

 Builder
    |
    |---- Register Builder Account
    |---- Create Project
    |---- Add Plot Inventory
    |---- Upload Legal Documents
    |---- Manage Plot Status

 Investor
    |
    |---- Search Plots
    |---- Filter & Compare Plots
    |---- View Legal Compliance
```

```
    |---- Book Plot
    |---- View Investment Analytics

 Admin
    |
    |---- Verify Builder
    |---- Approve Legal Documents
    |---- Monitor System Data
```

Purpose:
Helps stakeholders understand system functionality from a user perspective.

2. Sequence Diagram (Plot Booking Process)

Shows **step-by-step interaction between components** when a user books a plot.

```
Investor
   |
   | 1. Search Plot
   v
Frontend (React)
   |
   | 2. Request Plot Data
   v
Backend API (Node/Express)
   |
   | 3. Fetch Plot Data
   v
MongoDB Database
   |
   | 4. Return Plot Data
   v
Backend API
   |
   | 5. Send Plot Details
   v
Frontend
   |
   | 6. User Books Plot
   v
Backend API
   |
   | 7. Save Booking Record
   v
MongoDB
```

Purpose:
Explains how components communicate during booking operations.

3. Activity Diagram (Legal Compliance Verification)

This shows the **workflow of legal document verification.**

```
Start
  |
  v
Builder uploads legal documents
  |
  v
System stores documents
  |
  v
Admin reviews documents
  |
  +-----------+
  | Approved? |
  +-----------+
     |     |
    Yes    No
     |      |
     v      v
Legal status updated   Request correction
     |
     v
Legal data visible to investors
     |
     v
End
```

Purpose:
Shows decision flow and system workflow logic.

4. Component Diagram (System Components)

Shows **how major software components interact**.

```
        +---------------------+
        |   React Frontend    |
        | (User Interface)    |
        +----------+----------+
                   |
                   |
                   v
        +---------------------+
        |   API Controllers   |
        |  (Node + Express)   |
        +----------+----------+
                   |
                   |
                   v
        +---------------------+
        |    Service Layer    |
        | Business Logic      |
        +----------+----------+
                   |
                   |
                   v
        +---------------------+
        |   Database Layer    |
        |      MongoDB        |
        +----------+----------+
                   |
                   |
                   v
```

```
+---------------------+
| External Services   |
| Legal Data /        |
| Analytics Engine    |
+---------------------+
```

Purpose:
Shows software architecture components and their relationships.

15. Conclusion

The Smart Plot Investment Platform provides a scalable digital solution for real estate plot investment management. The system integrates builder project management, plot marketplace functionality, legal compliance verification, and investment analytics into a single unified platform.